

Ming Sun

Scala Developer | Functional Programming & Distributed Systems

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[LinkedIn](#) | [Website](#) | [GitHub](#)

Profile

Scala developer specialising in functional programming and distributed systems, with over 20 years of software development experience. Strong expertise in the functional Scala ecosystem, asynchronous processing and cloud-based backend services.

Technical Skills

Scala 3, Java, Cats, Cats Effect, FS2, ZIO, Kyo
Http4s, Tapir, Akka/Pekko, Apache Kafka, MongoDB, PostgreSQL
AWS, GCP, Kubernetes, Docker, Argo CD, Jenkins, GitOps, CI/CD

Professional Experience

HM Revenue & Customs (HMRC), UK

Jun 2021 - Present

Scala Developer (Contract)

- Develop and enhance Scala backend services across HMRC's portfolio of online services.
- Developed [lazysm2](#), a personal open-source terminal interface for HMRC's service manager, adopted by multiple teams to streamline local service management and debugging.
- Deliver technical presentations to colleagues, including replacing Shapeless type tagging with Scala 3 opaque types, covering type safety, interoperability and migration considerations.
- Improve service resilience by addressing downstream dependency failures and request timeouts.
- Resolve data consistency issues involving duplicate requests and concurrent updates.
- Investigate and resolve intermittent production failures, using application logs and monitoring to identify underlying causes.

Sky, Osterley, UK

Mar 2019 - Aug 2020

Scala Developer

- Built highly available services supporting customer registration and device activation across frontend applications and devices.
- Developed functional Scala services with Http4s, Akka HTTP, Cats, Cats Effect and Circe, using MongoDB and Kafka for persistence and messaging.
- Deployed and operated services on AWS and Kubernetes as part of a self-organising Agile team.

Telegraph Media Group, London, UK

Oct 2018 - Feb 2019

Platform Engineer

- Developed high-traffic Scala and Java platform services powering The Telegraph's web, mobile and third-party applications.
- Enhanced services including the Content API, User Preferences and Sitemap Service, and supported delivery to AWS ECS and GCP Kubernetes through Jenkins.

Earlier Experience

WorldFirst, London, UK

Jan 2017 - Aug 2018

Java / Scala Developer

- Helped replace a failing monolithic compliance system with well-defined microservices in a six-engineer team, using Java 8, Scala, Kafka, Akka, Docker and Kubernetes.
- Designed and delivered integration between the internal ledger and a third-party card transaction gateway, working within compliance requirements and service-level expectations.
- Worked closely with business stakeholders on solution design and helped establish performance-testing practices for critical services.
- Helped establish the company's first Java team, engineering practices and test frameworks; contributed to developer mentoring and a twice-weekly functional-programming guild.

NET-A-PORTER.COM, London, UK

Mar 2011 - Mar 2016

Java Developer

- Developed and maintained NET-A-PORTER and MR PORTER web applications using Java, Spring, JavaScript, Node.js and MySQL within Agile product teams.
- Introduced BDD automation with Cucumber and Selenium and coached QA colleagues to write executable Gherkin scenarios.
- Replaced cron-based data flows with ActiveMQ, contributed to an authentication API, and improved product-search performance with Solr.
- Collaborated with the Shanghai team to integrate China UnionPay for the Chinese market.

Animation 2000 Ltd, Pinewood Studios, UK

Apr 2004 - Feb 2011

Java Developer

- Designed and developed a specialist film-industry payroll platform covering payroll calculations, invoicing, Sage exports and Barclays payment files.
- Implemented the HMRC Gateway submission protocol using Java XML processing, secure multithreaded HTTP communication and digital signatures with Bouncy Castle and Apache XML Security.
- Improved operational resilience through standby database setup and automated backup and restore procedures.

Education

MSc Computing Technologies

2003

Northumbria University, Newcastle

Technical Writing

Selected topics: Monads vs Algebraic Effects; Implicits in Scala 3; Free Monad to Tagless Final.

[Read on website](#)